



Rating Distortion and Structural Bias in the COPRAS Multi-Criteria Decision-Making Method: A Critical Reassessment

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Abstract: The reliability and interpretability of the Complex Proportional Assessment (COPRAS) method in multi-criteria decision-making (MCDM) have been critically re-evaluated. Although COPRAS has frequently been promoted as a method capable of separately assessing the influence of benefit and cost criteria without requiring explicit inversion of cost attributes, it is demonstrated that these claims are not fully supported by the mathematical structure of the method. A theoretical analysis reveals that COPRAS inherently relies on Sum normalization, through which hidden attribute prioritization and rating distortion may be introduced. Furthermore, it is shown that, in the presence of a single cost criterion, COPRAS becomes mathematically equivalent to the Weighted Sum Model (WSM) implemented with Sum normalization and a nonlinear inverse-sum transformation of cost criteria. Consequently, the purported methodological distinction between COPRAS and conventional additive aggregation approaches is substantially reduced. Particular attention is drawn to the nonlinear inversion embedded in the COPRAS formulation for cost criteria aggregation. Because the inverse transformation is applied to the total contribution of cost criteria rather than to individual criterion values, the resulting influence of cost attributes on the final utility score is shown to be only indirectly represented. Under certain conditions, significant discrepancies are produced between the nominal and actual contribution of cost criteria, thereby affecting both rating stability and ranking consistency. Through numerical demonstrations and comparative analyses, distortions in alternative ratings and rank reversals are identified when COPRAS results are compared with those obtained from the conventional WSM framework. The analysis further indicates that the observed inconsistencies are primarily associated with the combined effects of Sum normalization and nonlinear cost treatment. To address these limitations, the WSM integrated with a linear cost-transformation procedure based on the Reverse Sorting (ReS) algorithm is proposed as a more transparent and mathematically consistent alternative. The findings suggest that the application of COPRAS in practical MCDM problems should be approached with caution, particularly in decision environments where ranking sensitivity and interpretability are of critical importance.

Keywords: Multi-criteria decision-making; Complex Proportional Assessment; Weighted Sum Mode; Reverse Sorting algorithm; Relative performance indicator

1 Introduction

The problem of multi-criteria decision-making (MCDM) on a discrete set of alternatives can be delineated as follows [1]: there exist m objects (alternatives) A_i , each characterized by n attributes (features). The attributes are quantitative assessments (a_{ij}) of the selected (important) properties of the objects. For each attribute, a criterion C_j is defined to determine the most preferable value of the feature. The objective is to make an optimal choice from the discrete set of alternatives. This choice problem is complicated by conflicts among criteria, where one alternative may outperform others in one (or several) features while being inferior in others.

The integration of solutions derived from multiple models—often referred to as the Multi-Method Model (3M)—constitutes a contemporary trend in the field of MCDM. It suffices to refer to the comprehensive reviews [2, 3] and systematic discussions of current challenges [4] dedicated to this topic. The underlying assumption is that combining a wide range of methods enhances the reliability of the resulting decisions. This aligns with the majority

principle frequently employed in collective decision-making. Nevertheless, historical experience indicates that this principle does not invariably yield the best outcome among all available alternatives.

One of the important tasks of solution synthesis is the selection of an independent set of models. Recent studies have shown that a number of “new” methods duplicate known methods. In particular, using the Relative Performance Indicator (RPI), six identical methods for aggregating individual attributes of alternatives were established [4, 5]: Weighted Sum Model (WSM), Multi-Attributive Border Approximation Area Comparison (MABAC), Technique for Order Preference by Similarity to Ideal Solution with L_1 metric (TOPSIS), Ratio System (RS) approach, Multi-Attribute Ideal-Real Comparative Analysis (MAIRCA) and Ranking of Alternatives with Weights of Criterion (RAWEC), provided that each model uses the same linear normalization method.

Another emerging trend is the development of hybrid models that integrate the most effective components of individual procedures from different methodologies, such as normalization/inversion, weighting, and the subsequent aggregation of alternatives attributes. Typical examples include the Max normalization method, weight estimation using the Analytic Hierarchy Process, Pairwise Comparisons and EigenVector method, and the TOPSIS aggregation method. To substantiate their effectiveness, researchers often compare the outcomes generated by these hybrid models with those obtained from established MCDM methods, including Value Measurement Methods or Goal/Reference Level Models. Such comparisons typically employ Spearman’s rank correlation coefficient for ranked lists and Pearson’s correlation coefficient for rating lists. In the absence of formal model comparison criteria, many authors also invoke consistency with well-established MCDM approaches, together with general theoretical arguments and principles, as supporting evidence for the proposed methods [6].

In this study, within the framework of the 3M approach, a comprehensive analysis of the effectiveness of the Complex Proportional Assessment (COPRAS) method was performed using quantitative methods. The COPRAS method, introduced by Zavadskas et al. [7], belongs to the class of value measurement methods and has numerous examples of implementation in solving multi-criteria selection problems in various application areas. In the COPRAS method, the influence of maximizing and minimizing criteria on the evaluation result is considered separately and the main advantage of COPRAS, according to the authors and their numerous followers, is that COPRAS does not apply the inversion of cost criteria.

It is important to note that neither the original descriptions of the method nor subsequent studies provide explicit indicators of its effectiveness. Such indicators could include, for instance, a high discriminative power; stability of rankings (or ratings) with respect to minor variations in the input data; a reduced number of subjective parameters compared to other methods; or the capability to address specific classes of problems that distinguish it from existing analogues. A simple comparison with the WSM method shows that the COPRAS method is essentially one of the forms of additive aggregation of normalized and weighted attributes for each alternative. This study shows that this form was chosen extremely poorly and is inferior to the WSM method in all respects. The strict framework of the method should also be considered a weakness. Data normalization, inversion, weighting of criteria are standard MCDM methods, independent of the aggregation method and they are easy to combine. If the authors include specific procedures for the COPRAS method, then it should be explained why this is the case and why another version of the method is not acceptable. This also applies, in particular, to most of the MCDM methods presented in the literature.

A recent review of the effectiveness of COPRAS summed up almost 30 years of its use and characterizes COPRAS as an outstanding method [8]. The wide range of applications and the number of citations is impressive. Both of these indicators can be easily explained. First, the method must be declared effective, and specialists in applied areas (considering it innovative) will “raise the flag high” in terms of application and citations. However, the above-mentioned review does not contain any scientific arguments for the effectiveness or innovativeness of the COPRAS method. What advantages does the COPRAS method have, according to the same review? It turns out (Table 1) that the listed advantages (there are 4 of them) are in no way related to the effectiveness of the method.

The limitations of the COPRAS method discussed in [8], although aimed at clarifying its underlying nature, are not unique to COPRAS itself; rather, they reflect general constraints inherent in MCDM methods. Likewise, common extensions of COPRAS designed to handle interval data and fuzzy evaluations [9, 10] are not intrinsically linked to the aggregation formula itself.

A large number of citations and applications of COPRAS may also be due to the fact that researchers are attracted by the intricate final formula (see Section 2.5, Eq. (24)). It seems that the complex formula includes more information, reflects more relationships. And this formula inspires more confidence than simple additive summation in WSM. In this study, we will try to explain the essence of the formula and show its incorrectness.

Podvezko [11] compares the Simple Additive Weighting (SAW) method, which is closely related to WSM, with the COPRAS method. However, the comparison is limited in scope because it emphasizes the presumed advantages of COPRAS without providing sufficient theoretical or quantitative justification. In particular, Podvezko [11] argues that SAW requires cost criteria to be transformed into benefit criteria before aggregation, whereas COPRAS avoids this limitation by treating benefit and cost criteria separately. This argument is further associated with the assumption

that the separate quantitative assessment of benefit- and cost-criterion contributions can improve the interpretation of the final alternative ratings and reveal possible instability in the evaluation results. Nevertheless, as shown in the present study, this interpretation requires further mathematical scrutiny, because COPRAS implicitly performs a nonlinear transformation of the cost-criterion contribution.

Table 1. Advantages of the COPRAS method and comments

Advantages According to Previous Study [8]	Comments (Our Research)
Simplicity and ease of understanding, making it accessible to users with varying levels of expertise.	There is no explanation of what the simplicity is or what the ease is, making it accessible to users with varying levels of expertise.
Ability to handle both quantitative and qualitative criteria, facilitating comprehensive decision-making.	Methods for joint processing of quantitative and qualitative criteria are not included in the final COPRAS aggregation formula.
Incorporates the relative importance of criteria, allowing for customized weighting based on decision-maker preferences.	The relative importance of criteria is used in all Multiple Criteria Decision Making (MCDM) models. This is a standard approach when solving multi-criteria problems.
Effective for ranking and selecting alternatives in diverse fields, demonstrating versatility and applicability.	The COPRAS method solves the same class of multi-criteria problems as its analogues. There are no performance indicators for the COPRAS method.

In this study, we examine not only the effectiveness of the COPRAS method but also the rationale and motivation behind its use. It is demonstrated that, similar to the WSM/SAW, COPRAS employs inversion for cost criteria; however, the inversion is implemented through implicit formulas expressed as nonlinear fractional rational functions. Rather than improving the estimates, this approach significantly distorts the original data when compared with WSM. Consequently, COPRAS transforms the attribute values of the alternatives in a nonlinear manner, altering the distances between values in a way that is not proportional to the original scale and thereby distorting the information about the objects. Rankings of the alternatives are then derived from this distorted information base.

The paper has the following structure. The second section contains a brief description of the theoretical information necessary for conducting the quantitative and qualitative analysis regarding the MCDM ranking model, the method for comparing the ranking of two or more MCDM methods based on RPI. The main methods of normalization/inversion are given in Section 2.3. The third section identifies all the problematic aspects of the COPRAS method. A rigorous proof of the equivalence of COPRAS to the WSM method is given for the case of one cost criterion (Appendix A). COPRAS contains a sum method of normalization in its structure, which results in data bias and a hidden priority of individual attributes. COPRAS implicitly contains a nonlinear inversion of criteria in the form of the inverse value of the total contribution of cost criteria. The fourth section presents a numerical example of COPRAS implementation using various linear normalization methods and a comparison of the results with the ranking results based on WSM.

2 Methodology

This section contains a brief description of the theoretical information required for conducting quantitative and qualitative analysis regarding the MCDM ranking model, the method for comparing the ranking of two or more MCDM methods based on RPI, the basic principles of data normalization and inversion.

2.1 The Multiple-Criteria Decision-Making Ranking Model

The rank model of MCDM [5, 12] is a mapping of a discrete set of attributes of alternatives, defined by the decision matrix $D[m \times n]$, into a one-dimensional set $Q[m \times 1]$ —ratings of alternatives:

$$F^{(k)} : D \rightarrow Q_i^{(k)} \quad (1)$$

The rating list Q is sorted in descending (or ascending) order, based on which a ranking list of alternatives is formed. Without loss of generality, we will assume that the sorting and renumbering of these two lists is performed in the form:

$$Q_1 \geq Q_2 \geq \dots \geq Q_m \quad (2)$$

$$A_1 \succ A_2 \succ \dots \succ A_m \quad (3)$$

where, $\succ$ denotes preference, and the subscripts determine the rank of the alternative.

Each method $F^{(k)}$ includes aggregation of normalized attributes of alternatives (x_{ij}), taking into account the significance (w_j) of the criteria, and for methods based on the distance from the ideal (target level models), the models use different distance metrics, denoted by dm :

$$Q^{(k)} = F^{(k)}(x_{ij}, \omega_j, dm) \quad (4)$$

The differences in the aggregation Eq. (4) lead to one of three types of models:

(M1)—Value Measurement methods, such as WSM:

$$Q_i = \sum_{j=1}^n w_j x_{ij} \quad (5)$$

(M2)—Goal or Reference Level Models, such as TOPSIS [1];

(M3)—Outranking Techniques, such as PROMETHEE (Preference Ranking Organization Method for Enrichment of Evaluations) [13].

2.2 Multi-Criteria and Aggregation of Individual Attributes

In MCDM problems, there are three types of criteria: benefit criteria, for which the goal is larger-the-better (LTB), cost criteria, for which the goal is smaller-the-better (STB), and criteria for which the target is the best value (TTB), for which the optimal value is within the data interval and is specified by the context of the problem. If the methodology for solving a multi-criteria problem is reduced to the convolution of criteria, then this predetermines the transformation of all criteria to the goal of either “max” or “min”.

To align the objectives of the criteria, data inversion is used. Inversion is defined as a rearrangement of a sorted list in reverse order [14]. For MCDM models (M1) based on “Value Measurement”, with a common goal of LTB, cost criteria are inverted, and with STB, benefit criteria are inverted. For criteria with a set target value, with the general goal of the LTB task, the inversion is performed for all values greater than the target, and with the goal of STB, the inversion is performed for all values less than the target [15].

For the target level model (M2), the criterion is based on the proximity to the ideal solution. The objectives are matched by determining the distance from the ideal (best) values for the benefit criteria, and the distance from the anti-ideal (worst values) for the cost criteria. This allows one to minimize (or maximize) a linear combination of these distances. In the (M2) model, no data inversion is required, but the criteria in $F^{(k)}$ are matched by the goal of maximizing the distance from the anti-ideal, or, conversely, by minimizing the distance from the ideal.

For model (M3), data inversion is also present, but implicitly, through the choice of preference functions.

2.3 Normalization and Inversion

The convolution of criteria for a multi-criteria problem is performed only for normalized data. Normalization of multivariate data not only brings data to a dimensionless form, but also matches the normalization scales. This is a necessary element of transforming multivariate data. A detailed study of normalization and inversion of multivariate data is presented in the study [15]. Since the convolution of criteria of multivariate data is performed only for normalized data, the inversion procedure must be consistent with the normalization procedure.

A significant part of MCDM models use a nonlinear inversion of the decision matrix of the form $1/a$. This inversion is often used as a component of such models as, for example, WSM, Weighted Product Model (WPM), Weighted Aggregated Sum Product Assessment (WASPAS) [16], COPRAS [7], Additive Ratio Assessment (ARAS) [17], COmbinative Distance-based ASsessment (CODAS) [18], TOPSIS [1], etc. Exactly the same approach is used in objective methods for assessing the weight of criteria based on the information contained in the decision matrix: Entropy weight method—EWM [19], Method based on the Removal Effects of Criteria (MEREC) [20], Simultaneous Evaluation of Criteria and Alternatives (SECA) [21], method of Criterion Impact LOSs (CILOS) [22], etc.

It is clear that a monotone strictly decreasing function $1/a$ will preserve the sequence of values in the list and invert the data, so that larger values become smaller and vice versa. In the study [14] it is shown that such an inversion is incorrect. A nonlinear transformation changes the data, or rather changes the distances between the values. And this means that the normalized data are fundamentally distorted compared to the original (natural) values. As an alternative, a linear inversion method is proposed—Reverse Sorting (ReS) algorithm, which eliminates all problems of nonlinear transformation.

The main linear normalization methods for multivariate data required for this study are summarized as follows [15]:

For benefit criteria (C^+) and cost criteria (C^-), the general form of linear normalization is given as by Eq. (6).

$$\begin{aligned} x_{ij} &= \text{Norm}(a_{ij}) = \frac{a_{ij} - a_j^*}{k_j}, \quad \text{for } j \in C^+; \\ \bar{x}_{ij} &= \frac{1/a_{ij}}{t_j}, \quad \text{for } j \in C^-. \end{aligned} \quad (6)$$

The Max normalization method and its corresponding nonlinear inversion procedure, iMax, are defined in Eq. (7).

$$\begin{aligned} x_{ij} &= \text{Max}(a_{ij}) = \frac{a_{ij}}{k_j}, \quad a_j^* = 0, \quad k_j = a_j^{\max}, \quad \text{for } j \in C_j^+; \\ t_j &= \frac{1}{a_j^{\min}}, \quad \text{for } j \in C_j^-. \end{aligned} \quad (7)$$

The Sum normalization method and its corresponding nonlinear inversion procedure, iSum, are given in Eq. (8).

$$\begin{aligned} x_{ij} &= \text{Sum}(a_{ij}) = \frac{a_{ij}}{k_j}, \quad a_j^* = 0, \quad k_j = \sum_{i=1}^m a_{ij}, \quad \text{for } j \in C_j^+; \\ t_j &= \sum_{i=1}^m \left(\frac{1}{a_{ij}} \right), \quad \text{for } j \in C_j^-. \end{aligned} \quad (8)$$

Vector normalization and its corresponding nonlinear inversion procedure, iVec, are expressed in Eq. (9).

$$\begin{aligned} x_{ij} &= \text{Vec}(a_{ij}) = \frac{a_{ij}}{k_j}, \quad a_j^* = 0, \quad k_j = \sqrt{\frac{1}{m} \sum_{i=1}^m a_{ij}^2}, \quad \text{for } j \in C_j^+; \\ t_j &= \sqrt{\frac{1}{m} \sum_{i=1}^m \left(\frac{1}{a_{ij}} \right)^2}, \quad \text{for } j \in C_j^-. \end{aligned} \quad (9)$$

The ReS algorithm provides a universal linear inversion procedure for cost or benefit criteria across all normalization methods, as shown in Eq. (10).

$$\begin{aligned} x_{ij} &= \text{Norm}(a_{ij}), \quad \forall j, \\ \bar{x}_{ij} &= \text{ReS}(x_{ij}) = -x_{ij} + x_j^{\min} + x_j^{\max}, \quad \text{for } j \in C_j^-. \end{aligned} \quad (10)$$

The Max–Min normalization procedure is defined in Eq. (11).

$$\begin{aligned} x_{ij} &= \text{Max-Min}(a_{ij}), \quad a_j^* = a_j^{\min}, \quad k_j = a_j^{\max} - a_j^{\min}, \quad \text{for } j \in C_j^+; \\ \bar{x}_{ij} &= \frac{a_j^{\max} - a_{ij}}{k_j} = \text{ReS}(x_{ij}), \quad \text{for } j \in C_j^-. \end{aligned} \quad (11)$$

The z -score normalization method transformed to the interval $[0, 1]$ is given by Eq. (12) and Eq. (13).

$$u_{ij} = \frac{a_{ij} - a_j^*}{k_j}, \quad a_j^* = \text{mean}(a_{ij}), \quad k_j = \sqrt{\frac{1}{m} \sum_{i=1}^m (a_{ij} - a_j^*)^2}. \quad (12)$$

$$x_{ij} = \frac{u_{ij} - u_j^{\min}}{u_j^{\max} - u_j^{\min}}, \quad \forall i, j, \quad \bar{x}_{ij} = \text{ReS}(x_{ij}), \quad \text{for } j \in C_j^-. \quad (13)$$

The IZ transformation maps normalized values to the interval $[I, Z]$ as shown in Eq. (14).

$$x_{ij} = \frac{a_{ij} - a_j^{\min}}{a_j^{\max} - a_j^{\min}}(Z - I) + I = \text{Max-Min}(a)(Z - I) + I, \quad [I, Z] \subset [0, 1], \quad \forall i, j. \quad (14)$$

For example, when the transformation is applied to the interval $[md, 1]$, it can be written as Eq. (15).

$$I = md = \text{median} \left\{ \min_i [\text{Max}(a_{ij})] \right\}, \quad Z = 1. \quad (15)$$

where, $\text{Max}(a_{ij})$ denotes the value obtained by applying the Max normalization method in Eq. (7) to a_{ij} .

According to the study [15], multivariate normalization methods are based on three main principles:

- (i) normalized values preserve information about the original data;
- (ii) normalization should be invariant to scaling of measurement scales;
- (iii) no single attribute should have priority during subsequent joint processing of dimensionless features.

Since the attributes of objects and the ranges of their natural values differ greatly from each other, each of the features uses its own scale. Therefore, requirement (i) for multivariate normalization is effectively impossible to satisfy. In this case, some violation of the data structure (mutual distances) occurs and there is a shift in the domains of normalized values, which also determines the priorities of individual attributes. Despite this, this approach is considered generally accepted.

Requirement (ii) is satisfied only for linear transformations. Dispositions are determined for a fixed j by Eq. (16):

$$dx_{ij} = \frac{|x_{i+1,j} - x_{ij}|}{x_j^{\max} - x_j^{\min}}, \quad i = 1, \dots, m-1, \quad \forall j \quad (16)$$

The study [15] showed that dispositions are invariant with respect to data scaling. In particular, dispositions will not depend on the normalization coefficients k_j and t_j in Eqs. (6)–(13). Given that the ReS inversion is a linear transformation, requirements (ii) for data inversion are also met.

The third principle (iii), namely the absence of priority among attributes, cannot be fully satisfied in multivariate normalization. In multi-criteria problems, linear normalization methods usually produce anisotropic scaling because each attribute is normalized according to its own scale, which may be inconsistent with the scales of other attributes. As a result, the ranges of normalized values may differ across attributes, and the corresponding domains of normalized values may shift relative to one another. According to the classification of multivariate normalization methods [15], partial consistency can be achieved through the following variants:

- (1) alignment of the largest normalized values;
- (2) alignment of the average normalized values;
- (3) alignment of both the largest and smallest normalized values;
- (4) alignment of the average normalized values and standard deviations; and
- (5) normalization of deviations from the ideal or anti-ideal values based on variants (2)–(4).

When nonlinear normalization procedures are applied, the dispositions of the original data are violated. The same problem arises in nonlinear inversion procedures of the $1/a$ type.

As a result, the normalized data may be structurally distorted relative to the original values. To address this problem, the ReS algorithm has been proposed as a linear inversion method that avoids the distortions caused by nonlinear transformations [14].

Figure 1 illustrates the violation of requirements (i) and (ii) when the Sum normalization method is combined with the nonlinear iSum inversion procedure defined in Eq. (8), which is based on the $1/a$ transformation. In this example, the decision problem is to select the best alternative from eight alternatives, each characterized by five attributes. The decision matrix is given in Eq. (17), where the fourth criterion is treated as a cost criterion and is therefore inverted to make it consistent with the LTB objective.

$$D = \begin{pmatrix} 71 & 4500 & 150 & 1056 & 478 \\ 85 & 5800 & 145 & 2680 & 564 \\ 76 & 5600 & 135 & 1230 & 620 \\ 74 & 4200 & 160 & 1480 & 448 \\ 82 & 6200 & 183 & 1350 & 615 \\ 81 & 6000 & 173 & 1565 & 580 \\ 80 & 5900 & 160 & 1650 & 610 \\ 85 & 4700 & 140 & 1750 & 667 \end{pmatrix} \quad (17)$$

In subgraph (A) of Figure 1 shows the normalized values of the matrix A using the sum method in Eq. (8). The domains of the normalized values have different ranges and there is a shift in the domains relative to each other, which contradicts the principle (i). In subgraph (B) of Figure 1 presents the inverted normalized values of the fourth attribute for all alternatives. Specifically, the fourth column of the decision matrix in Eq. (17) is normalized and inverted using the iSum procedure, and the results are compared with those obtained using the linear ReS inversion in Eq. (10). For the iSum method, the domain of the inverted normalized values is shifted relative to the domain of the Sum-normalized values for the same attribute. This shift changes the contribution of the inverted values to the overall performance indicator of the alternatives. A similar shift occurs when the Vec/iVec normalization–inversion procedure defined in Eq. (9) is used. By contrast, under ReS inversion, the domains of the normalized and inverted values coincide. Similarly, for the Max/iMax procedure defined in Eq. (7), the domains of the normalized and inverted values also coincide.

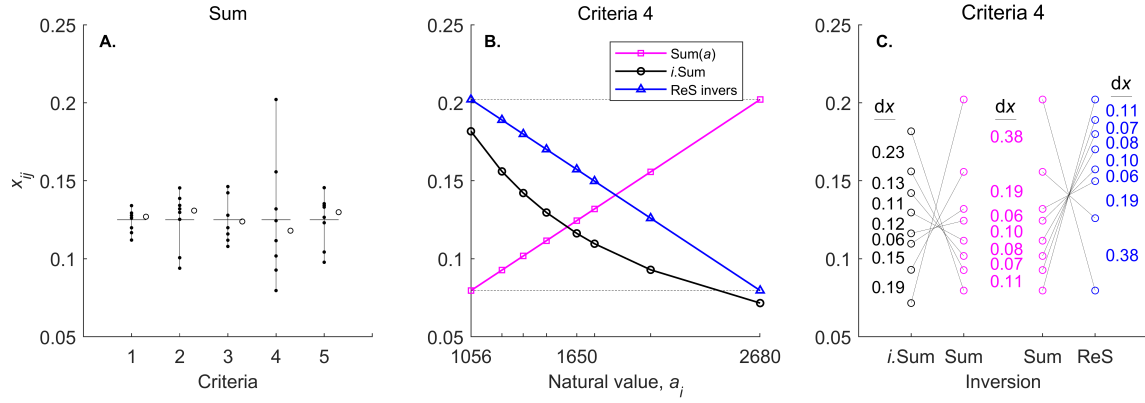


Figure 1. Domain shifts and disposition distortions caused by nonlinear iSum inversion for the fourth attribute: (A) domains of Sum-normalized values; (B) comparison of normalized and inverted values for the fourth attribute; and (C) comparison of dispositions for the fourth attribute

Subgraph (C) of Figure 1 shows the dispositions (dx_{i4}) of the normalized values of the fourth attribute of all alternatives. These dispositions are calculated for the iSum inversion method ($1/a$) and the ReS inversion method and are compared with the dispositions of the original Sum-normalized values. Under iSum inversion, the dispositions do not correspond to those of the original Sum-normalized values. For some alternatives, the dispositions differ by up to 1.7 times, indicating a significant distortion of the original data structure. This distortion occurs because nonlinear inversion does not preserve central symmetry, as illustrated in subgraph (C) in Figure 1. Similar disposition distortions may also occur when iMax inversion in Eq. (7) or iVec inversion in Eq. (9) is used.

2.4 Comparison of Multiple-Criteria Decision-Making Ranking Models

The presence of a large number of MCDM methods requires the development of a tool that can be used to compare them with each other and to synthesize solutions. Comparison of the ranks of different methods is simple, but is rather rough, since the ranks do not reveal the degree of superiority of the alternatives among themselves. A rating list, in contrast to a rank list, reflects the “thin” structure of relationships between alternatives [4]. However, the ratings of different methods are defined in different scales determined by the method of aggregating partial values of attributes.

As a result of applying two different MCDM ranking models, two rating lists are obtained: $Q_i^{(1)}$ and $Q_i^{(2)}$. The lower index corresponds to the i th alternative, the upper index corresponds to the model number.

Definition 1: Two rating lists are equivalent if they are transformed into each other using a linear transformation. For comparison, it is rational to perform an ordering of the rating of alternatives and transform both lists to the form of a RPI [4]:

$$\Delta Q_p = \frac{(Q_p - Q_{p+1})}{rng(Q)} \times 100\%, \quad p = 1, \dots, m-1 \quad (18)$$

where, Q_p is the value of the performance indicator corresponding to the alternative ranked p -th in the ordered list, and $rng(Q) = \max_i Q_i - \min_i Q_i = Q_1 - Q_m$.

In this case, the indicator ΔQ is a relative (given in the Q scale) increase or decrease in the performance indicator for the ordered list of alternatives and the property is satisfied:

$$\sum_{p=1}^{m-1} \Delta Q_p = 100 \quad (19)$$

Definition 2: Two rating lists are equivalent if they have the same RPI:

$$\Delta Q_i^{(1)} = \Delta Q_i^{(2)}, \quad \forall i = 1, \dots, m-1 \quad (20)$$

Property: If two rating lists are equivalent, then both ranking models are equivalent.

Thus, ΔQ_i indicator serves as an important characteristic for comparing MCDM models. In particular, in the study [4] using RPI, six methods of aggregating individual attributes of alternatives with identical results were established: WSM, MABAC, TOPSIS, RS, MAIRCA and RAWEC, provided that each model uses the same linear normalization method and the same criterion weights.

2.5 Complex Proportional Assessment

The step-by-step algorithm of the COPRAS method is as follows [7]:

Step 1. Define m alternatives and n criteria. The decision matrix is expressed as $\mathbf{D} = (a_{ij})[m \times n]$;

Step 2. Define the criterion weights $\mathbf{w} = (w_j)[1 \times n]$;

Step 3. Normalize the decision matrix using the Sum normalization method:

$$x_{ij} = \text{Norm}(a_{ij}) = \text{Sum}(a_{ij}) = \frac{a_{ij}}{\sum_{i=1}^m a_{ij}}, \quad i = 1, \dots, m, j = 1, \dots, n \quad (21)$$

Step 4. Determine the separate contributions of the benefit and cost criteria. If the first g criteria are benefit criteria and the remaining criteria are cost criteria, the contribution of the benefit criteria and cost criteria are calculated as Eq. (22) and Eq. (23).

$$S_{+i} = \sum_{j=1}^g w_j \cdot x_{ij}, \quad j \in C_j^+ \quad (22)$$

$$S_{-i} = \sum_{j=g+1}^n w_j \cdot x_{ij}, \quad j \in C_j^- \quad (23)$$

Step 5. Calculate the COPRAS performance indicator Q_i for each alternative:

$$Q_i = S_{+i} + \sum_{i=1}^m S_{-i} / \left(S_{-i} \cdot \sum_{i=1}^m \frac{1}{S_{-i}} \right) \quad (24)$$

where, Q_i is the final performance indicator of the i -th alternative. A larger value of Q_i indicates a more preferable alternative.

3 Detailed analysis of the Complex Proportional Assessment method

Problem X: To illustrate the features of the COPRAS method and illustrate the main theoretical principles, this study defines a test problem of multicriteria choice (problem X from eight alternatives, each of which is defined by a set of 5 attributes in the context of 5 criteria. The decision matrix D is given by Eq. (17). The data for each attribute are randomly generated values regardless of the distribution law. To analyze the COPRAS method, it is necessary to include both benefit and cost criteria in the problem, given that the COPRAS method separately processes benefit and cost criteria. In this problem, the second, fourth, and fifth attributes represent cost attributes. The weighting coefficients of the criteria are chosen arbitrarily, $w = (1 \ 2 \ 4 \ 3 \ 1)$, but differently for greater generality. The importance of the attributes (criteria weights) is determined randomly. Random assignment is determined by the fact that in this problem we do not study the effect of changing the importance of criteria on the final decision. This eliminates the risk of erroneous assessment of significance. When solving the MCDM problem, the weights are normalized by dividing by their sum.

3.1 Analysis of Normalization Method Used in Complex Proportional Assessment

In the COPRAS method, the sum method of normalization is used in step 3. The studies [4, 15] show that there are no restrictions on use another similar normalization method. In this case, the ratings of the alternatives change, and in some cases the ranks change as well. Figure 2 shows the domains of normalized values for problem X obtained using the Sum method in Eq. (8), the Max method in Eq. (7), and the IZ[md,1] transformation in Eq. (14) and Eq. (15).

It is evident that for the Max and IZ normalization methods, the spread of normalized values differs from the Sum method. As shown in subgraph (B) and (C) in Figure 2, they are the same in the maximum value, but different in the sum. Thus, COPRAS leads to different results depending on the normalization method. Using the COPRAS method, the ratings and ranks of the alternatives of the problem X were calculated. In Figure 2, the alternatives with the first three ranks and the RPI (ΔQ) of the alternatives for the ordered ranking list are additionally presented. Recall that for COPRAS there is no need to invert the values of the cost criteria during normalization.

The study [15] showed that the Sum normalization method (like the Vec method) leads to a shift in the domains of the normalized values of various attributes and produces a different range. As a result, some of the criteria will receive priority for contributing to the integral indicator even before assigning a weight.

It is easy to see in subgraph (A) in Figure 2 that for the sum method, the 4th attribute has the greatest shift relative to others. The shift for the Max method is also significant, but the “lower” values, which contribute less to the integral rating, are shifted. The closer the value is to 1 (the share of the best), the higher its contribution to the rating. The IZ[md,1] normalization is a linear transformation for Max normalization [15], in which the smallest

values for all criteria are aligned. The smallest value is the median of the smallest values of all attributes for the Max method. This is an analogue of the max-min normalization method (Eq. (11)), but differs in the exclusion of zero values. For the max-min normalization method, the smallest normalized value is always 0. If any of the alternatives takes the smallest values for all cost criteria, then $S_{-I} = 0$ for it. Then, in Eq. (24), division by 0 occurs. That is, the method cannot be implemented for max-min normalization. In particular, this problem for the max-min normalization method always occurs for the case when there is only one cost criterion in the problem.

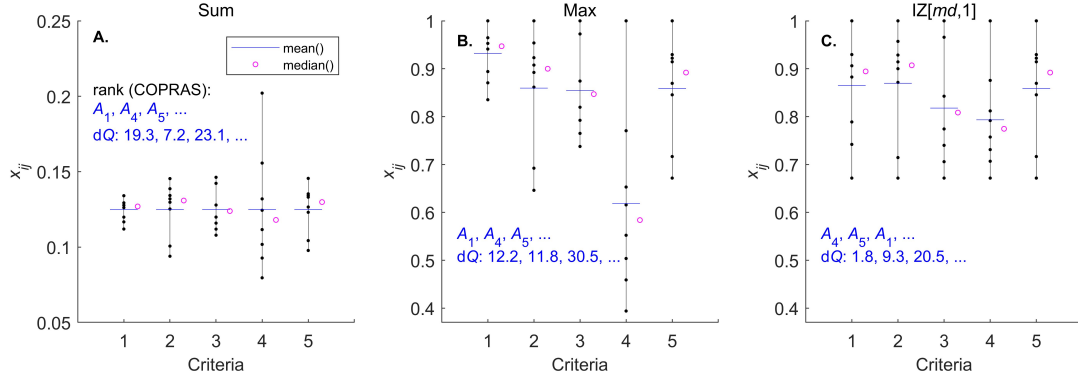


Figure 2. Domains of normalized values obtained by the Sum, Max and IZ[md,1] methods

If, when used in COPRAS with Sum and Max normalization, the ranks coincide and the ratings differ insignificantly, then when using IZ, the ranks of the alternatives also change. However, we do not draw deep conclusions on this basis. The fact is that the influence of the normalization method is obvious and is manifested for all methods, including WSM. But we want to emphasize that Sum is not the best of the methods, since there is always a shift in the domains of normalized values of individual attributes relative to each other. The shift occurs for both the largest and smallest values.

The interpretation of the results is important during normalization, since during aggregation we operate with shares of natural values. For the Sum method, in many cases, the characteristic scale of normalization of the j th attribute (k_j) as a sum of natural values does not make sense in terms of interpreting this result. In this case, the sum can be divided by m (the number of alternatives) and then the arithmetic mean value ($\bar{a}_j$) is obtained:

$$\bar{a}_j = k_j / m$$

Note that for a linear or homogeneous aggregation function in Eq. (4), scaling all attributes by dividing by m will not affect the ranking result. The rating is scaled, but the RPI value will not change. Therefore, the normalized values of the sum method can be interpreted as proportions of the average:

$$x_{ij} = \frac{a_{ij}}{m \cdot \bar{a}_j} = \frac{a_{ij}}{\sum_{i=1}^m a_{ij}}.$$

This means that the more the attribute value exceeds the average value, the higher its contribution to the rating. But it is easy to see that the number of normalized values exceeding the average and the relative proportions of these values do not depend on the normalization method if it is linear. This is also seen in Figure 2. Therefore, the determining factor is how far these values are from the average. Therefore, the shift in the domains of various criteria is the reason for the priority of individual criteria. For the Max normalization method, the largest values of all attributes are the same and the priority of the criteria at the normalization stage (before assigning the weight of the criteria) is levelled. This means that the Sum method of normalization is inferior to the Max method of normalization.

3.2 Comments on the comparison of Complex Proportional Assessment and Simple Additive Weighting

This section provides some comments on the study [11] comparing COPRAS and SAW methods:

1) For SAW in the specified study, the Max/iMax normalization (Eq. (7)) is used, and for COPRAS, the Sum normalization method (Eq. (8)) is used. The use of different methods seems methodologically incorrect, since, in this case, not only the contribution of cost criteria to the integrated assessment of alternatives differs, but the contribution of benefit criteria will also differ. It also seems incorrect to conclude that the COPRAS method does not perform inversion of cost attributes. We will further show in Section 3.3–3.5 based on mathematical transformations that this does not correspond to the actual calculation formula of the COPRAS method.

2) In addition, the author provides obvious properties:

$$\begin{aligned} S_+ &= \sum_{i=1}^m S_{+i} = \sum_{j=1}^g w_j \\ S_- &= \sum_{i=1}^m S_{-i} = \sum_{j=g+1}^n w_j \end{aligned} \quad (25)$$

but it does not explain whether they are relevant to the comparison of SAW and COPRAS. S_{+i} , S_{-i} represent the share of the contribution of the benefit (cost) criteria to the rating of the i th alternative. If these ratings are added up for all alternatives, questions arise as to how to interpret this and how and for what purpose to apply? The article does not provide an interpretation that answers these questions.

The first comment emphasizes that the MCDM method represents the unity and synthesis of the method of normalizing the decision matrix, the method of determining the priority of criteria, and the method of aggregating particular features to the integrated assessment of alternatives.

3.3 Separate evaluation of criteria in the Complex Proportional Assessment method

Many authors, including the authors of the COPRAS method, believe that taking into account the separate contribution of the benefit and cost criteria in the COPRAS method allows for a better analysis of the results for decision-making. This is highly questionable. For any MCDM model, the integral rating includes all the attributes of the alternatives selected for analysis. If you find, for example, that the contribution of the benefit attributes is greater, for example, 3:1, what follows from this? What decisions should the decision makers make?

The second objection is as follows. For any multi-criteria optimization problem, the criteria are explicitly or implicitly reduced to one type, for example, LTB. This procedure can be called “inversion”, or “goal reversal”. In TOPSIS, for example, the inversion for the cost criteria is defined through removal from the anti-ideal. In fact, the optimization problem is transformed and there are no criteria of different types in it. Therefore, the authors will not be able to isolate the separate contribution of the cost criteria. In fact, the result is determined by the correctness of the inversion procedure. This is the contribution of the inverted data.

3.4 Inversion in the Complex Proportional Assessment method

Many authors, including the authors of the COPRAS method, believe that COPRAS does not use the inversion of cost criteria. Here we will show that this is not the case. Moreover, the iSum inversion used in COPRAS, as shown above in Section 2.3, distorts the dispositions of the source data. Let us prove that COPRAS uses the inversion of cost criteria. First, we note that the value in Eq. (24) equals to a constant.

$$\frac{\sum_{i=1}^m S_{-i}}{\sum_{i=1}^m \frac{1}{S_{-i}}} = C = \text{Const.} \quad (26)$$

Then the calculation formula for the alternatives' rating has a simple form:

$$Q_i = S_{+i} + \frac{C}{S_{-i}} \quad (27)$$

Eq. (27) is a compact final form of the COPRAS method. The first term for Q_i is the contribution of the benefit criteria to the overall rating, the second term is the contribution of the inverted values of the cost criteria to the overall rating.

We will show that the second term is comparable, and sometimes equal, to the contribution for the inverted values of the cost attributes of the WSM method. The WSM method, as well as COPRAS, can be represented as the sum of two terms R_{+i} and R_{-i} , reflecting the contribution of benefit and cost criteria.

$$Q_i = R_{+i} + R_{-i} = \sum_{j=1}^g w_j \cdot x_{ij} + \sum_{j=g+1}^n w_j \cdot \bar{x}_{ij} \quad (28)$$

where, $x_{ij} = a_{ij}/k_j$ are the normalized values of the benefit attributes, $\bar{x}_{ij}$ are the inverted values of the cost attributes.

Following Eq. (22) and Eq. (28), $R_{+i} = S_{+i}$. If there are no cost criteria in the problem ($R_{-i} = S_i = 0$), then $WSM \equiv COPRAS$ when applying the same procedure for normalizing the decision matrix.

To compare the WSM methods with COPRAS, it is necessary to apply the same method of normalizing the decision matrix as in the COPRAS method, i.e., we use the normalization method $x_{ij} = \text{Sum}(a_{ij})$ and use the

corresponding nonlinear inversion $\bar{x}_{ij} = (1/a_{ij})/t_j$ (Eq. (8)). In the presence of cost criteria, the difference between the two methods is determined by the value of the ratio:

$$u_i = \frac{C}{S_{-i}}/R_{-i} \quad (29)$$

The contribution of cost attributes to the rating of the i th alternative for the COPRAS method is:

$$\frac{C}{S_{-i}} = \frac{C}{\sum_{j=g+1}^n w_j \times x_{ij}} = \frac{C}{\sum_{j=g+1}^n w_j \times \frac{a_{ij}}{k_j}} \quad (30)$$

The contribution of cost attributes to the rating of the i th alternative for the WSM method using the nonlinear inversion $\bar{x}_{ij} = (1/a_{ij})/t_j$, is equal to:

$$R_{-i} = \sum_{j=g+1}^n w_j \times \bar{x}_{ij} = \sum_{j=g+1}^n w_j \times \frac{1/t_j}{a_{ij}} \quad (31)$$

It is easy to see that in COPRAS, in accordance with Eq. (30), the inversion is defined as the inverse value of the weighted sum of the normalized values of the cost attributes, and WSM each term in Eq. (31) is inverted. These two values differ both upwards and downwards. The result is determined by the initial data and the number of cost criteria. In one particular case, this contribution coincides. Namely, when there is only one cost criterion, and the WSM method uses the Sum/iSum normalization. The proof is presented in the Appendix.

Thus, for the Sum normalization with one cost criterion, the COPRAS method is equivalent to WSM if the same Sum normalization and the nonlinear inversion of iSum in Eq. (8) for the cost criteria were used for WSM. This means that the second term is exactly equal to the contribution for the inverse values of the cost attributes. In fact, COPRAS uses the nonlinear inversion of iSum, contrary to the assertion of the authors of the method.

If Max or Vec normalization methods are used for WSM, and Sum normalization for the COPRAS method, then R_{+i} and S_{+i} , will obviously differ, and the analysis of the contribution of cost attributes becomes meaningless.

Studies show that for rank MCDM models the choice of normalization is not unambiguous [4, 5, 15]. This choice is determined only by the requirements for the normalized values. Namely, during normalization it is necessary to preserve the information of natural values—preserve the dispositions between the attribute values and exclude the priority of normalized values of individual attributes over others—exclude the shift of domains of normalized values. This applies equally to inversion. The first principle is satisfied by all linear normalization methods. Therefore, nonlinear inversion is incorrect. The second principle is achieved for the Max–Min method and its analogues with IZ transformation of normalized values.

For comparison, for the problem X defined above, the average value of the indicator u was calculated in accordance with Eq. (29) for a different number k of cost criteria (sequentially and starting from the last criterion, $g = n \setminus k$) and different normalization methods (Table 2).

With one cost criterion and Sum/iSum normalization (see also the rigorous proof in the Appendix), COPRAS is equivalent to WSM.

Table 2. Mean ratio ($\bar{u}$) and standard deviation (σ_u) of cost-criterion contributions between Complex Proportional Assessment (COPRAS) and Weighted Sum Model (WSM) for Problem X

Number of Cost Criteria, k	1	2	3	4	5
	$\bar{u} \pm \sigma_u$				
Sum/iSum	1 ± 0	1.00 ± 0.017	1.00 ± 0.024	1.00 ± 0.019	1.00 ± 0.017
Max/iMax	1.08 ± 0.000	0.95 ± 0.016	0.97 ± 0.019	1.00 ± 0.015	1.00 ± 0.013
Vec/iVec	1.00 ± 0.012	0.99 ± 0.017	1.00 ± 0.023	1.00 ± 0.018	1.00 ± 0.017
IZ[md, 1]/Reverse Sorting	1.06 ± 0.016	0.94 ± 0.012	0.95 ± 0.007	0.97 ± 0.006	0.98 ± 0.006

Note: k denotes the number of criteria treated as cost criteria. For $n = 5$, the cost criteria are selected sequentially from the last criterion, with $g = n - k$ benefit criteria.

In the calculations for both COPRAS and WSM the same normalizations were used. The difference in the contribution of the cost criteria is insignificant, about 2%, and this value does not provide any new information. This difference may affect the result, given the high sensitivity of the solution to the variation of the initial data. But we will never know the exact reason, due to the lack of a truth criterion. This indicates that COPRAS is just a copy of WSM with a more complex Eq. (24).

3.5 Case with Multiple Cost Criteria

To examine the case with multiple cost criteria, consider first a setting with two cost criteria. In this case, the analytical transformations presented in the Appendix for the single-cost-criterion case can no longer be applied, because no common factor can be isolated from the corresponding cost-criterion terms.

For COPRAS, the contribution of the cost criteria can be expressed as follows:

$$Q_i = S_{+i} + \frac{C}{S_{-i}} = S_{+i} + \frac{C}{w_{n-1} \times \frac{a_{i,n-1}}{k_{n-1}} + w_n \times \frac{a_{i,n}}{k_n}} \quad (32)$$

For comparison, the corresponding contribution in WSM is given by:

$$Q_i = S_{+i} + w_{n-1} \times \frac{1}{t_{n-1} \times a_{i,n-1}} + w_n \times \frac{1}{t_n \times a_{i,n}} \quad (33)$$

Since w_j , t_j and k_j are constants for $j = n - 1$ and $j = n$, the difference between the COPRAS and WSM contributions depends on the following terms respectively:

$$\frac{1}{\alpha \times a_{i,n-1} + \beta \cdot a_{i,n}} \quad \text{and} \quad \frac{b}{a_{i,n-1}} + \frac{d}{a_{i,n}}$$

In the single-cost-criterion case, the two contributions reduce to a simple proportional relationship. When more than one cost criterion is involved, the comparison is instead between the inverse of a weighted linear combination and a weighted linear combination of inverse values. These quantities are analogous to a weighted arithmetic mean (wAM) and a weighted harmonic mean (wHM), respectively. Although the harmonic mean is generally no greater than the arithmetic mean under equal weights, the coefficients in Eq. (33) make the two quantities numerically close in the present setting.

For problem X, when three cost criteria are considered for eight alternatives, the ratios of the COPRAS cost-criterion contribution to the corresponding WSM contribution are as follows:

$$u = (1.0034 \quad 1.0136 \quad 1.0071 \quad 0.9605 \quad 0.9925 \quad 1.0045 \quad 0.9990 \quad 1.0144)$$

These values indicate that, depending on the alternative, the cost-criterion contribution in COPRAS may be slightly greater or smaller than the corresponding contribution in WSM. However, in this example, the differences remain small.

4 Numerical Comparison of Complex Proportional Assessment and Weighted Sum Model

4.1 Ranking Results under Different Normalization Methods

Using numerical analysis, we examine how the ranking results based on WSM change when COPRAS is used with different normalization methods. Table 3 presents the results of ranking the alternatives for the problem X defined above in Section 3, using WSM and COPRAS and varying the normalization. In addition to the normalization and inversion procedures defined in Section 2.3, the dSum normalization procedure is used for comparison. It is defined as follows:

$$\begin{aligned} x_{ij} &= (a_{ij} - a_j^*) / k_j \\ k_j &= \sum_{i=1}^m (a_j^{\max} - a_{ij}) \\ a_j^* &= a_j^{\max} - k_j \\ a_j^{\max} &= \max(a_{ij}) \end{aligned} \quad (34)$$

This example shows that when integrating a linear normalization method other than “Sum” into COPRAS, the results are similar to those of the WSM method with the same normalization method. The first four options use normalization methods without bias, and the ranking and rating results are in good agreement. The next three methods are normalization methods with bias. In this case, the ranking and rating results are also in perfect agreement. The ΔQ value shows how much the ratings of alternatives with adjacent rankings differ. This important indicator helps guide decisions in situations where the ratings of alternatives differ only slightly.

COPRAS does not provide any new additional information different from the WSM results.

Thus, this example demonstrates that the ranking results of the COPRAS method and various WSM models with different normalization/inversion methods yield comparable results. The results of solving the test problem, as shown in Table 3, indicate that COPRAS is a constructive analogue of WSM, distorting, among other things, the data

relationships. COPRAS produces both a change in rating and, in some cases, a change in the rank of alternatives, due to the algorithm's nonlinear transformation of the contribution of cost attributes to the overall ranking of alternatives. The equivalence of COPRAS and WSM in the case of a single cost criterion, and the slight difference in results from the WSM model in Section 3.4 for the case of two or more cost criteria, confirm that the COPRAS algorithm, in terms of the contribution of cost criteria, is evaluative. The results presented in Table 3 for this example show that the difference is insignificant: the value of $\bar{u}$ characterizing the degree of difference is of the order of 1. The difference in ranks 1–2 occurs for the normalization/inversion of IZ[md, 1]/ReS and Z[0, 1]/ReS. In the latter case, the difference in the contribution of cost criteria is significant ($\bar{u} = 0.8$).

Table 3. Ranking results for problem X using Weighted Sum Model (WSM) and Complex Proportional Assessment (COPRAS) under different normalization methods

rank	Sum/iSum				Max/ReS				Sum/ReS				Vec/ReS				dSum/ReS				IZ[md, 1]/ReS				Z[0, 1]/ReS			
	WSM		COPRAS		WSM		COPRAS		WSM		COPRAS		WSM		COPRAS		WSM		COPRAS		WSM		COPRAS		WSM		COPRAS	
	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ	#A	ΔQ
1	1	–	1	–	1	–	1	–	1	–	1	–	1	–	1	–	4	–	4	–	5	–	4	–	5	–	4	–
2	4	21.2	4	19.3	4	3.6	4	12.2	2	4.0	4	9.3	4	8.5	4	18.6	5	10.0	5	15.8	4	4.7	5	1.8	4	3.3	4	26.3
3	5	5.8	5	7.2	5	5.4	5	11.8	5	2.4	5	7.2	5	2.6	5	7.6	1	9.0	1	1.7	1	13.7	1	9.3	1	11.2	5	19.7
4	3	23.8	3	23.1	7	33.3	6	30.5	3	25.1	3	23.1	3	26.2	3	24.1	6	15.0	6	20.3	6	12.1	6	20.5	6	17.3	6	14.4
5	7	10.5	7	6.1	6	2.0	7	1.3	7	4.6	7	6.1	7	3.9	7	5.3	7	19.0	7	19.2	7	17.7	7	20.0	7	14.9	7	15.9
6	6	2.1	6	3.6	3	0.0	3	1.1	6	7.5	6	3.6	6	6.9	6	3.1	3	28.8	3	28.0	3	28.5	3	29.6	3	24.8	3	20.4
7	8	23.7	8	20.9	8	31.2	8	28.4	8	19.5	8	20.9	8	20.6	8	21.9	2	11.2	2	8.0	8	18.6	8	16.8	8	20.6	2	1.2
8	2	12.9	2	19.8	2	24.5	2	14.8	2	32.1	2	19.8	2	31.4	2	19.3	8	7.0	8	6.9	2	4.6	2	2.0	2	7.8	8	2.1
$\bar{u} \pm \sigma_u$	1.00 $\pm$ 0.019				0.94 $\pm$ 0.029				0.92 $\pm$ 0.033				0.92 $\pm$ 0.033				1.00 $\pm$ 0.006				0.98 $\pm$ 0.011				0.80 $\pm$ 0.268			

Note: #A denotes the alternative number at each rank position, ordered from best to worst. ΔQ denotes the relative distance between two adjacent alternatives in the ordered rating list, as defined in Section 2.4; the values sum to 100%. $\bar{u} \pm \sigma_u$ denotes the mean and standard deviation of the ratio u defined in Eq. (29). ReS = Reverse Sorting. The normalization/inversion procedures are defined in Eqs. (7)–(15) and Eq. (34).

4.2 Rank Deviations between Complex Proportional Assessment and Weighted Sum Model

Assuming that WSM provides the reference ranking, a numerical simulation was conducted to evaluate the rank deviations produced by COPRAS. As discussed in the previous section, the ranking results depend on the structure of the decision matrix. Therefore, a set of simulated MCDM problems with the same dimensions as the original test problem was generated.

The original decision matrix D , defined in Section 3, has dimensions 8×5 . The second, fourth, and fifth criteria are treated as cost criteria. Based on this matrix, a set of simulated decision matrices $D^* = (a_{ij}^*)$ was generated. For each criterion j , the generated values were sampled from the interval $[a_j^{\min}, a_j^{\max}]$, where $a_j^{\min}$ and $a_j^{\max}$ denote the minimum and maximum values of criterion j in the original decision matrix, respectively. Each element of D^* was generated as follows:

$$a_{ij}^* = a_j^{\min} + (a_j^{\max} - a_j^{\min}) r_{ij}, \quad r_{ij} \sim \mathcal{U}(0, 1), \quad i = 1, \dots, m, \quad j = 1, \dots, n.$$

where, r_{ij} is a random variable uniformly distributed over the interval $[0, 1]$. The generated matrices can be regarded as alternative realizations of the same decision-making problem, with the same criterion ranges but different sets of alternatives. For each simulated decision matrix, rankings were obtained using WSM with Sum/iSum normalization and COPRAS with Sum normalization, both under the same criterion-weight vector w . A total of $N = 10,000$ were performed. Rank-1 alternatives did not match (regardless of other ranks) in 8% of problems; rank-2 alternatives did not match in 15% of problems; and the ranks of at least one alternative did not match in 38% of problems.

Critical to decision-making is the discrepancy between rank 1 alternatives, which accounted for approximately 8% of the total number of generated problems.

In most generated problems, COPRAS and WSM produced the same first- and second-ranked alternatives. Such consistency may be interpreted by proponents of COPRAS as evidence of its practical agreement with WSM, particularly because COPRAS is often claimed to avoid the transformation of cost criteria. However, as shown in Section 3.4, COPRAS effectively transforms the contribution of cost criteria through a weighted harmonic-mean structure. Therefore, the observed mismatches—approximately 8% for the first-ranked alternatives and 15% for the second-ranked alternatives—remain decision-relevant, as they may affect the final selection or the priority order of alternatives.

Although the simulation results are specific to problem X and to the data-generation procedure used here, they demonstrate that rank deviations may occur between COPRAS and WSM. This finding indicates that the apparent agreement between the two methods does not eliminate the risk of rank changes caused by the implicit treatment of cost criteria in COPRAS.

5 Conclusions

The tendency to synthesize solutions based on a large number of MCDM models is based on the assumption that each method includes different relationships between alternatives, which increases the overall information content and improves the reliability of the result. This premise suggests defining a list of acceptable methods for solving a specific problem. These methods should be independent and reflect different relationships between alternatives. The presence of a large number of MCDM methods requires research that would compare them with each other. In particular, recent studies have identified six methods for aggregating private attributes of alternatives that are identical in results: WSM, MABAC, TOPSIS, RS, MAIRCA and RAWEC.

Another problem is to identify effective MCDM methods in a large arsenal. Efficiency indicators may include, for example, high resolution; rating (or rank) stability to small variations in the source data; or to determine that a method includes fewer subjective parameters than another method; or indicate that the method differs from its analogues in that it allows solving a certain class of problems, etc.

Within the framework of such a task, this paper provides a comprehensive analysis of the effectiveness of the COPRAS method. The ability to analyse the separate influence of maximizing and minimizing criteria on the assessment result, as declared in COPRAS, is not true. The declared advantage that COPRAS does not apply the inversion of cost criteria is also not true.

The COPRAS method uses Sum normalization, which results in data bias and the hidden priority of individual attributes. It is shown that for a single cost criterion, COPRAS is equivalent to the WSM model with Sum/iSum normalization/inversion. This equivalence is achieved when WSM employs neither the best normalization method nor the best cost criteria inversion method. The COPRAS calculation formula contains a nonlinear inversion of the total contribution of cost criteria. This nonlinear inversion distorts the original information. Therefore, the contribution of cost criteria to the overall ranking is estimated and, in some cases, differs significantly from the actual contribution. In other cases, the deviations in estimates from the actual contribution are not so large, and therefore the COPRAS method is comparable to WSM in terms of results. However, COPRAS does not provide any additional decision-making information, there is no interpretation of the contribution of cost attributes to the overall ranking of an alternative, and the calculation formula is more complex than that of WSM.

Thus, COPRAS can be characterized as an inferior analogue of the WSM method. Given all these concerns, we do not recommend using the COPRAS method for solving MCDM problems to avoid potentially misleading results. The authors specifically draw attention to the error in using nonlinear inversion of cost criteria in MCDM methods. The universal (applicable to all normalization methods) ReS algorithm for linear inversion has been described in detail in the scientific literature since 2019.

The directions of further research are determined by the development of the MCDM model synthesis approach. Specifically, future research should assess the effectiveness of methods used to solve the MCDM problem from the standpoint of increasing the information content and reliability of hybrid models.

Author Contributions

Conceptualization, I.M. and I.Z.; methodology, I.M. and I.Z.; software, I.M.; validation, I.Z.; formal analysis, I.Z.; investigation, I.Z.; resources, I.Z.; data curation, I.M.; writing—original draft preparation, I.Z.; writing—review and editing, I.M.; visualization, I.M.; supervision, I.Z.; project administration, I.Z.; funding acquisition, I.Z. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data (decision matrix) supporting our research results are included within the article or supplementary material.

Conflicts of Interest

The authors declare no conflict of interest..

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Appendix

Inversion of cost criteria for COPRAS and WSM. Special case $g = n - 1$, i.e., cost criterion is one.

When $g = n - 1$, Eqs. (28) and (29) are transformed to the form:

$$\frac{C}{S_{-i}} = \frac{C \cdot k_n}{w_n \cdot a_{in}}$$

$$R_{-i} = \frac{w_n}{t_n \cdot a_{in}}$$

The difference between the two methods is determined by the value of the ratio u of these two quantities.

$$u = \frac{C \cdot k_n}{w_n} : \frac{w_n}{t_n} = \frac{C \cdot k_n \cdot t_n}{w_n^2} \quad (A1)$$

The constant C is defined below:

$$C = \frac{\sum_{i=1}^m S_{-i}}{\sum_{i=1}^m \frac{1}{S_{-i}}} = \frac{\sum_{i=1}^m w_n \cdot x_{in}}{\sum_{i=1}^m \frac{1}{w_n \cdot x_{in}}} = \frac{w_n^2 \sum_{i=1}^m x_{in}}{\sum_{i=1}^m \frac{1}{x_{in}}} = \frac{w_n^2 \sum_{i=1}^m a_{in}/k_n}{\sum_{i=1}^m \frac{k_n}{a_{in}}} = \frac{w_n^2 \sum_{i=1}^m a_{in}}{k_n^2 \sum_{i=1}^m \frac{1}{a_{in}}} \quad (A2)$$

Substituting the expression for C into Eq. (A1) transforms Eq. (A1) into Eq. (A3):

$$u = \frac{C \cdot k_n \cdot t_n}{w_n^2} = \frac{t_n \cdot \sum_{i=1}^m a_{in}}{k_n \cdot \sum_{i=1}^m \frac{1}{a_{in}}} \quad (A3)$$

Eq. (A3) does not contain an explicit normalization formula and can therefore be applied to unbiased linear normalization methods, including Max, Sum, and Vec, together with their corresponding nonlinear inversion procedures, iMax, iSum, and iVec, as defined in Eqs. (7)–(9).

For the Sum/iSum normalization–inversion procedure, substituting the corresponding scale factors t_n and k_n into Eq. (A3) gives:

$$u^{(1)} = \frac{t_n \cdot \sum_{i=1}^m a_{in}}{k_n \cdot \sum_{i=1}^m \frac{1}{a_{in}}} = \frac{\sum_{i=1}^m \frac{1}{a_{in}} \cdot \sum_{i=1}^m a_{in}}{\sum_{i=1}^m a_{in} \cdot \sum_{i=1}^m \frac{1}{a_{in}}} = 1 \quad (A4)$$

For the Max/iMax normalization–inversion procedure, the inverted normalized values are defined as $\bar{x}_{ij} = \frac{a_j^{\min}}{a_{ij}} = \frac{1/a_{ij}}{1/a_j^{\min}}$ and $t_j = \frac{1}{a_j^{\min}}$.

Substituting these values into Eq. (A3) gives:

$$u^{(2)} = \frac{\frac{1}{a_n^{\min}} \cdot \sum_{i=1}^m a_{in}}{a_n^{\max} \cdot \sum_{i=1}^m \frac{1}{a_{in}}} = \frac{\sum_{i=1}^m x_{in}}{\sum_{i=1}^m \bar{x}_{in}} \quad (A5)$$

An analogous result is obtained for the Vec/iVec normalization–inversion procedure. The inverted normalized values are defined as follows:

$$\bar{x}_{ij} = \frac{1/a_{ij}}{\left(\sum_{i=1}^m (1/a_{ij})^2 \right)^{0.5}}$$

$$t_j = \left(\sum_{i=1}^m (1/a_{ij})^2 \right)^{0.5}$$

Substituting these values into Eq. (A3) gives:

$$u^{(3)} = \frac{t_n \cdot \sum_{i=1}^m a_{in}}{k_n \cdot \sum_{i=1}^m \frac{1}{a_{in}}} = \frac{\sqrt{\sum_{i=1}^m (1/a_{in})^2} \cdot \sum_{i=1}^m a_{in}}{\sqrt{\sum_{i=1}^m (a_{in})^2} \cdot \sum_{i=1}^m (1/a_{in})} \quad (A6)$$

Thus, under the Sum/iSum normalization–inversion procedure with a single cost criterion, COPRAS is mathematically equivalent to WSM when WSM applies the corresponding nonlinear iSum inversion to the cost criterion.